



AI Playbook and Toolkit for Public Health Departments

Record Linkage, Entity Resolution, and Deduplication

OPS 210

Record linkage and entity resolution help public health systems determine whether records refer to the same person, provider, facility, address, organization, event, or outbreak. AI tools often rely on linked data even when the linkage process is invisible to users. This module teaches learners how matching and deduplication affect AI outputs, surveillance quality, and equity.

Level	applied/foundational
Primary track	Operations and Data Quality Track
Appears in tracks	technical-architecture

Learning Objectives

- Explain why record linkage matters for public health AI.
- Distinguish between deterministic matching, probabilistic matching, entity resolution, and deduplication.
- Identify risks from false matches and missed matches.
- Describe how thresholds, clerical review, and data quality affect matching.
- Produce a linkage risk review for an AI-supported workflow.

Module Overview

Record linkage and entity resolution help public health systems determine whether records refer to the same person, provider, facility, address, organization, event, or outbreak. AI tools often rely on linked data even when the linkage process is invisible to users. This module teaches learners how matching and deduplication affect AI outputs, surveillance quality, and equity.

Why This Topic Matters for Public Health AI

Record linkage matters because public health work often depends on combining information across systems. A surveillance system may need to associate a lab report with an existing case. An IIS may need to match vaccination records from multiple providers. A vital records linkage may support maternal and child health analysis. An AI tool using these linked records may summarize history, predict risk, classify cases, or identify duplicates. If linkage is wrong, the AI output may be wrong.

False matches and missed matches have different consequences. A false match can combine information from different people or events, creating privacy, safety, and accuracy risks. A missed match can split one person or event into multiple records, leading to duplicate work, incomplete histories, or undercounting. Both errors can affect public health decisions.

Linkage can also create equity risks. Names, addresses, phone numbers, language, housing status, migration patterns, and documentation practices vary across communities. Matching rules that work well for one population may perform poorly for another. AI workflows that rely on linked data should therefore include subgroup and equity review.

Core Concepts

Record linkage. Record linkage is the process of determining whether records from one or more sources refer to the same entity. The entity may be a person, provider, facility, address, organization, event, or outbreak.

Entity resolution. Entity resolution is the broader process of identifying, matching, merging, separating, and managing records about the same entity. It includes matching rules, thresholds, review processes, and stewardship.

Deterministic matching. Deterministic matching uses exact or rule-based agreement across selected fields. It is easier to explain but may miss matches when fields are incomplete or inconsistent.

Probabilistic matching. Probabilistic matching estimates the likelihood that records refer to the same entity even when fields differ. It can improve recall but requires thresholds, validation, and review.

Clerical review. Clerical review is human review of uncertain matches. It is essential when automatic matching is not reliable enough for the downstream use.

Public Health Example

An AI-supported case summary may pull together lab reports, case notes, hospitalization information, and prior reports for the same person. If the person matching process incorrectly combines two people, the summary may include information that belongs to someone else. If the process misses a prior record, the summary may omit important history. The investigator needs visibility into linkage confidence and source records.

An outbreak detection workflow may link complaints, lab results, facility inspections, and geographic data. Entity resolution may be needed to identify whether different facility names refer to the same establishment. AI-generated cluster summaries can be misleading if the facility resolution process is weak.

Technical Deep Dive

Matching design begins with the entity being matched and the decision that depends on the match. Person matching may use name, date of birth, address, phone, medical record number, email, sex, or other identifiers. Facility matching may use name, address, license number, geocode, and ownership information. Event matching may use dates, locations, condition, exposure, and related records.

Thresholds determine when records are automatically linked, automatically separated, or sent for review. Low thresholds may increase false matches. High thresholds may increase missed matches. The right balance depends on the use case. A workflow that affects individual services may need a different tolerance for error than an aggregate trend analysis.

AI readiness requires documentation of matching methods, fields, thresholds, review processes, error rates, and known limitations. If an AI tool uses linked records, staff should know whether linkage is deterministic, probabilistic, manual, or hybrid. They should also know whether uncertain matches are visible to users and whether AI outputs indicate linkage uncertainty.

Risks, Failure Modes, and Guardrails

False matches. False matches can combine data from different people or entities. Guardrails include conservative thresholds for consequential uses, clerical review, and source display.

Missed matches. Missed matches can fragment histories and create duplicate work. Guardrails include matching quality monitoring, feedback loops, and review queues.

Equity variation. Matching may perform differently across populations. Guardrails include subgroup evaluation, alternative identifiers, and review of naming and address patterns.

Hidden linkage uncertainty. AI outputs may present linked data as certain. Guardrails include confidence indicators, source records, and user review of uncertain matches.

Application to AI-Supported Workflows

Generative AI may summarize linked records, so source attribution and linkage confidence should be visible. Predictive analytics may use features created from linked records, making match quality part of model validity. NLP may extract identifiers from text to support linkage, which requires privacy and accuracy review. Agentic AI should not automatically merge or update records without strong approval and rollback procedures.

Reflection Questions

Which workflows in your agency depend on matching records across systems?

What fields are used for matching, and how complete are they?

Which populations may be more likely to experience missed or incorrect matches?

What happens when records are incorrectly merged or not merged?

Where should human review be required?

Practical Exercise

Create a linkage risk review for one AI-supported public health workflow. Include the matching purpose, data sources, matching fields, known field quality issues, match method, thresholds, clerical review process, false match risks, missed match risks, equity considerations, and AI output affected.

Expected Artifact or Evidence

Linkage risk review

Deduplication workflow map

Match review questions

Equity and escalation notes

Expected Assignment

- Linkage risk review
- Deduplication workflow map
- Match review questions
- Equity and escalation notes

Knowledge Check

- 1. What is a false match?
- 2. What is a missed match?
- 3. Why does record linkage matter for AI?
- 4. Who may need to review uncertain matches?
- 5. What is strong completion evidence for this module?

References and Resources

- Public health IIS and surveillance data quality resources
- CDC data modernization resources
- Health information exchange data quality materials
- Agency matching and deduplication procedures